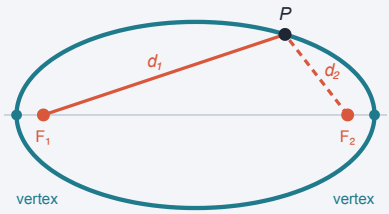
The Ellipse

College Algebra 2e, Example by Example
timelymath.github.io/college-algebra-examples

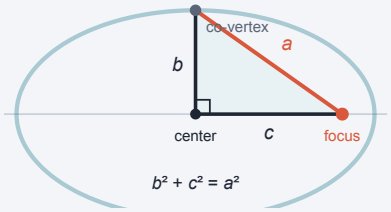
What an ellipse is

All the points whose distances to two fixed points, the **foci**, add to the same total: $d_1 + d_2 = 2a$.



Why $c^2 = a^2 - b^2$

From a co-vertex to a focus is exactly a, half of $2a$. That gives a right triangle with legs b and c.



Reading a, b and c

- a^2 is always the **larger denominator**, so $a > b$.
- The major axis runs along the variable that a^2 sits under: under x, the ellipse is wide; under y, it's tall.
- From the center, the vertices are a away, the co-vertices b away, and the foci c away, all along their axes.
- Find c with $c^2 = a^2 - b^2$, then take square roots: $a^2 = 25$ means the vertices are 5 away.

Standard forms, centered at (h, k)

	Major axis horizontal (wide)	Major axis vertical (tall)
EQUATION	$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$	$\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1$
AT THE ORIGIN	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$	$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$
VERTICES	$(h \pm a,\ k)$	$(h,\ k \pm a)$
CO-VERTICES	$(h,\ k \pm b)$	$(h \pm b,\ k)$
FOCI	$(h \pm c,\ k)$, where $c^2 = a^2 - b^2$	$(h,\ k \pm c)$, where $c^2 = a^2 - b^2$

General form to standard form (Example 6)

1. Group the x terms and the y terms; move the constant.
   $(4x^2 - 40x) + (9y^2 + 36y) = -100$
2. Factor out the coefficients of the squared terms.
   $4(x^2 - 10x) + 9(y^2 + 4y) = -100$
3. Complete each square. Add what you really added to the right side: $4 \cdot 25 = 100$ and $9 \cdot 4 = 36$.
   $4(x-5)^2 + 9(y+2)^2 = -100 + 100 + 36 = 36$
4. Divide so the right side is 1.
   $\frac{(x-5)^2}{9} + \frac{(y+2)^2}{4} = 1$, center $(5, -2)$, $a = 3$, $b = 2$

Common mistakes

- $c^2 = a^2 + b^2$ is the hyperbola. For an ellipse, $c^2 = a^2 - b^2$.
- **a^2 isn't always under x^2.** It's the larger denominator, wherever it is.
- **Signs of h and k:** $(x+2)^2$ means $h = -2$.
- **Balancing:** adding 25 inside $4(\ )$ adds 100, so add 100 to the right side.
- **Stopping early:** it isn't standard form until the right side is 1.

Watch the examples · scan or click; every page has a Try It with a hint and the answer

EXAMPLE 1 Writing the Equation of an Ellipse Centered at the Origin in Standard Form

EXAMPLE 2 Writing the Equation of an Ellipse Centered at a Point Other Than the Origin

EXAMPLE 3 Graphing an Ellipse Centered at the Origin

EXAMPLE 4 Graphing an Ellipse Centered at the Origin from an Equation Not in Standard Form

EXAMPLE 5 Graphing an Ellipse Centered at (h, k)

EXAMPLE 6 Graphing an Ellipse Centered at (h, k) by First Writing It in Standard Form

EXAMPLE 7 Locating the Foci of a Whispering Chamber

ALL OF 8.1 Every example, the section playlist and the review video

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